

## SAFETY DATA SHEET

Revision Date 01-Apr-2024

Revision Number 5

### 1. Identification

**Product Name** Nafion® D-521 dispersion, 5% w/w in water and 1-propanol

**Cat No. :** 42117

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |            |
|--------------------------------------------------|------------|
| Flammable liquids                                | Category 3 |
| Serious Eye Damage/Eye Irritation                | Category 1 |
| Specific target organ toxicity (single exposure) | Category 3 |
| Target Organs - Central nervous system (CNS).    |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Flammable liquid and vapor  
Causes serious eye damage  
May cause drowsiness or dizziness



### Precautionary Statements

#### Prevention

Wear protective gloves/protective clothing/eye protection/face protection

Avoid breathing dust/fume/gas/mist/vapors/spray

Use only outdoors or in a well-ventilated area

Keep away from heat/sparks/open flames/hot surfaces. - No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Use explosion-proof electrical/ventilating/lighting equipment

Use only non-sparking tools

Take precautionary measures against static discharge

Keep cool

#### Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Call a POISON CENTER or doctor/physician if you feel unwell

#### Skin

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

#### Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER or doctor/physician

#### Fire

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

#### Storage

Store in a well-ventilated place. Keep container tightly closed

Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

None identified

## 3. Composition/Information on Ingredients

| Component                                                                                                                                            | CAS No     | Weight % |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| n-Propyl alcohol                                                                                                                                     | 71-23-8    | 48.2     |
| Water                                                                                                                                                | 7732-18-5  | 45.2     |
| Ethanesulfonic acid,<br>2-[1-[difluoro[(trifluoroethenyl)oxy]methyl]-1,2,2,2-tetrafluoroethoxy]-1,1,2,2-tetrafluoro-, polymer with tetrafluoroethene | 31175-20-9 | 3.6      |
| Ethyl alcohol                                                                                                                                        | 64-17-5    | 3        |

## 4. First-aid measures

### General Advice

If symptoms persist, call a physician.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Most important symptoms and effects</b> | Causes severe eye damage. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting     |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

## 5. Fire-fighting measures

|                                         |                                                   |
|-----------------------------------------|---------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                          |
| <b>Flash Point</b>                      | 24 °C / 75.2 °F                                   |
| <b>Method -</b>                         | No information available                          |
| <b>Autoignition Temperature</b>         | No information available                          |
| <b>Explosion Limits</b>                 |                                                   |
| <b>Upper</b>                            | No data available                                 |
| <b>Lower</b>                            | No data available                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                          |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

None known.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 2                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system.                                                                                                                     |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.** Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component        | ACGIH TLV      | OSHA PEL                                                                                                                                                                         | NIOSH                                                                                                       | Mexico OEL (TWA) |
|------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------|
| n-Propyl alcohol | TWA: 100 ppm   | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 500 mg/m <sup>3</sup><br>(Vacated) STEL: 250 ppm<br>(Vacated) STEL: 625 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 500 mg/m <sup>3</sup> | IDLH: 800 ppm<br>TWA: 200 ppm<br>TWA: 500 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 625 mg/m <sup>3</sup> | TWA: 100 ppm     |
| Ethyl alcohol    | STEL: 1000 ppm | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup>                                                                 | IDLH: 3300 ppm<br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup>                                              | STEL: 1000 ppm   |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
 OSHA - Occupational Safety and Health Administration  
 NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                          |
|----------------------------------|--------------------------|
| Physical State                   | Liquid dispersion        |
| Appearance                       | Cloudy white             |
| Odor                             | No information available |
| Odor Threshold                   | No information available |
| pH                               | No information available |
| Melting Point/Range              | No data available        |
| Boiling Point/Range              | No information available |
| Flash Point                      | 24 °C / 75.2 °F          |
| Evaporation Rate                 | No information available |
| Flammability (solid,gas)         | Not applicable           |
| Flammability or explosive limits |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |

|                                        |                          |
|----------------------------------------|--------------------------|
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Density                                | 0.98                     |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                                                   |
|----------------------------------|-------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                        |
| Stability                        | Stable under normal conditions.                                   |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents                                           |
| Hazardous Decomposition Products | None under normal use conditions                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                          |
| Hazardous Reactions              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component        | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation              |
|------------------|---------------------------|------------------------------|------------------------------|
| n-Propyl alcohol | LD50 = 1870 mg/kg ( Rat ) | LD50 = 4049 mg/kg ( Rabbit ) | LC50 > 33.8 mg/L ( Rat ) 4 h |
| Water            | -                         | -                            | -                            |
| Ethyl alcohol    | LD50 = 7060 mg/kg ( Rat ) | Not listed                   | 20000 ppm/10H ( Rat )        |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                                                                                       | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| n-Propyl alcohol                                                                                                                | 71-23-8    | Not listed | Not listed | Not listed | Not listed | Not listed |
| Water                                                                                                                           | 7732-18-5  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Ethanesulfonic acid, 2-[1-[difluoro[(trifluoroethenyl)oxy]methyl]-1,2,2,2-tetrafluoroethoxy]-1,1,2,2-tetrafluoro-, polymer with | 31175-20-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |         |            |       |    |            |    |
|-------------------|---------|------------|-------|----|------------|----|
| tetrafluoroethene |         |            |       |    |            |    |
| Ethyl alcohol     | 64-17-5 | Not listed | Known | A3 | Not listed | A3 |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*NTP: (National Toxicity Program)*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*Mexico - Occupational Exposure Limits - Carcinogens*

*Mexico - Occupational Exposure Limits - Carcinogens*

*A1 - Confirmed Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Confirmed Animal Carcinogen*

*A4 - Not Classifiable as a Human Carcinogen*

*A5 - Not Suspected as a Human Carcinogen*

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Central nervous system (CNS)

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component        | Freshwater Algae                              | Freshwater Fish                                                  | Microtox                                                                                                                | Water Flea                                                                                         |
|------------------|-----------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| n-Propyl alcohol | Not listed                                    | Pimephales promelas:<br>LC50=4480 mg/L 96h                       | EC50 = 17700 mg/L 5 min<br>EC50 = 45000 mg/L 5 h<br>EC50 = 8686 mg/L 15 min<br>EC50 = 980 mg/L 12 h                     | EC50: 3339 - 3977 mg/L,<br>48h Static (Daphnia magna)<br>EC50: = 3642 mg/L, 48h<br>(Daphnia magna) |
| Ethyl alcohol    | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | Photobacterium<br>phosphoreum: EC50 = 34634<br>mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 = 35470<br>mg/L/5 min | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h                                                      |

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

| Component        | log Pow |
|------------------|---------|
| n-Propyl alcohol | 0.2     |
| Ethyl alcohol    | -0.32   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN1987  
Proper Shipping Name ALCOHOLS, N.O.S.  
Hazard Class 3  
Packing Group III

#### TDG

UN-No UN1987  
Proper Shipping Name ALCOHOLS, N.O.S.  
Hazard Class 3  
Packing Group III

#### IATA

UN-No UN1987  
Proper Shipping Name ALCOHOLS, N.O.S.  
Hazard Class 3  
Packing Group III

#### IMDG/IMO

UN-No UN1987  
Proper Shipping Name ALCOHOLS, N.O.S.  
Hazard Class 3  
Packing Group III

### 15. Regulatory information

#### United States of America Inventory

| Component                                                                                                                                           | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| n-Propyl alcohol                                                                                                                                    | 71-23-8    | X    | ACTIVE                                        | -                           |
| Water                                                                                                                                               | 7732-18-5  | X    | ACTIVE                                        | -                           |
| Ethanesulfonic acid, 2-[1-[difluoro[(trifluoroethenyl)oxy]methyl]-1,2,2,2-tetrafluoroethoxy]-1,1,1,2,2-tetrafluoro-, polymer with tetrafluoroethene | 31175-20-9 | X    | ACTIVE                                        | XU                          |
| Ethyl alcohol                                                                                                                                       | 64-17-5    | X    | ACTIVE                                        | -                           |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B))

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

China, X = listed, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS).

| Component                                                                                                                                         | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL        |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|------|-----------|-------|------|------|------|-------|-------------|
| n-Propyl alcohol                                                                                                                                  | 71-23-8    | X   | -    | 200-746-9 | X     | X    | X    | X    | X     | KE-29362    |
| Water                                                                                                                                             | 7732-18-5  | X   | -    | 231-791-2 | X     | X    |      | X    | X     | KE-35400    |
| Ethanesulfonic acid, 2-[1-[difluoro[(trifluoroethenyl)oxy]methyl]-1,2,2,2-tetrafluoroethoxy]-1,1,2,2-tetrafluoro-, polymer with tetrafluoroethene | 31175-20-9 | -   | X    | -         | -     | X    |      | -    | X     | 2009-3-4151 |
| Ethyl alcohol                                                                                                                                     | 64-17-5    | X   | -    | 200-578-6 | X     | X    | X    | X    | X     | KE-13217    |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations**

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product contains the following Proposition 65 chemicals.

| Component     | CAS No  | California Prop. 65                    | Prop 65 NSRL | Category                 |
|---------------|---------|----------------------------------------|--------------|--------------------------|
| Ethyl alcohol | 64-17-5 | Development (alcoholic beverages only) | -            | Developmental Carcinogen |

**U.S. State Right-to-Know Regulations**

| Component        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------------|---------------|------------|--------------|----------|--------------|
| n-Propyl alcohol | X             | X          | X            | -        | X            |
| Water            | -             | -          | X            | -        | -            |
| Ethyl alcohol    | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N



**U.S. Department of Homeland  
Security**

This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component                                                                                                                                                     | CAS No     | REACH (1907/2006) -<br>Annex XIV - Substances<br>Subject to Authorization | REACH (1907/2006) -<br>Annex XVII - Restrictions<br>on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 -<br>Candidate List of<br>Substances of Very High<br>Concern (SVHC) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| n-Propyl alcohol                                                                                                                                              | 71-23-8    | -                                                                         | Use restricted. See item<br>75.<br>(see link for restriction<br>details)               | -                                                                                                                 |
| Water                                                                                                                                                         | 7732-18-5  | -                                                                         | -                                                                                      | -                                                                                                                 |
| Ethanesulfonic acid,<br>2-[1-[difluoro[(trifluoroethyl)oxy]m<br>ethyl]-1,2,2,2-tetrafluoroethoxy]-1,1,<br>2,2-tetrafluoro-, polymer with<br>tetrafluoroethene | 31175-20-9 | -                                                                         | -                                                                                      | -                                                                                                                 |
| Ethyl alcohol                                                                                                                                                 | 64-17-5    | -                                                                         | -                                                                                      | -                                                                                                                 |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                                                                                                                          | CAS No     | OECD HPV       | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------|---------------------------------|------------------------------|--------------------------------------------------|
| n-Propyl alcohol                                                                                                                                   | 71-23-8    | Listed         | Not applicable                  | Not applicable               | Not applicable                                   |
| Water                                                                                                                                              | 7732-18-5  | Listed         | Not applicable                  | Not applicable               | Not applicable                                   |
| Ethanesulfonic acid,<br>2-[1-[difluoro[(trifluoroethyl)oxy]methyl]-1,2,2,2-tetrafluoroethoxy]-1,1,2,2-tetrafluoro-, polymer with tetrafluoroethene | 31175-20-9 | Not applicable | Not applicable                  | Not applicable               | Not applicable                                   |
| Ethyl alcohol                                                                                                                                      | 64-17-5    | Listed         | Not applicable                  | Not applicable               | Not applicable                                   |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

See table for values

| Component                                                                                                                                                                 | OECD PFAS | US (EPA) PFAS | EU (ECHA) PFAS | UK (HSE) PFAS | Chemsec PFAS (Sin<br>List) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------|---------------|----------------------------|
| Ethanesulfonic acid,<br>2-[1-[difluoro[(trifluoroethyl)oxy]methyl]-1,2,2,2-tetrafluoroethoxy]-1,1,2,2-tetrafluoro-, polymer with tetrafluoroethene<br>(CAS #: 31175-20-9) | Listed    | -             | Listed         | Listed        | -                          |

**PFAS Legend**

Listed = Meets the PFAS definition of the named authority

**Other International Regulations**

| Component | CAS No | Seveso III Directive | Seveso III Directive | Rotterdam | Basel Convention |
|-----------|--------|----------------------|----------------------|-----------|------------------|
|-----------|--------|----------------------|----------------------|-----------|------------------|

|                                                                                                                                                                 |            | (2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | (2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Convention (PIC) | (Hazardous Waste) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|-------------------|
| n-Propyl alcohol                                                                                                                                                | 71-23-8    | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |
| Water                                                                                                                                                           | 7732-18-5  | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |
| Ethanesulfonic acid,<br>2-[1-[difluoro[(trifluoroethenyl)<br>oxy]methyl]-1,2,2,2-tetrafluoro<br>ethoxy]-1,1,2,2-tetrafluoro-,<br>polymer with tetrafluoroethene | 31175-20-9 | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |
| Ethyl alcohol                                                                                                                                                   | 64-17-5    | Not applicable                                                                | Not applicable                                                               | Not applicable   | Annex I - Y42     |

## 16. Other information

**Prepared By**

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date**

01-Apr-2024

**Print Date**

01-Apr-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**